

The Contaminated City

Chief Scientific Officer • Riverton Contamination Response

15 DAYS • 61 STOPS • GRADE 12 • CONTAMCITY

THE OPENING CARD

A freight yard beside the river burned last night, and nobody can say what came off it. You are the city's Chief Scientific Officer, which means what gets measured — and what the city is told about it — is decided by you. Two neighbourhoods are downwind of a plume nobody has identified. The drinking-water intake for a hundred and forty thousand people sits nine hundred metres downstream of the outfall, and the utility has closed it, which means the city is drinking its reserves and the reserves are measured in days.

The Unknown Containers

BEFORE THE DAY — GREET

Six leads, one incident, and nobody has worked together before

The response was assembled on Tuesday out of whoever was reachable, and by Friday somebody will be signing a release based on a number a stranger produced. Okonjo, the analytical chemistry lead, wants you known to as many of the leads as you can before the first sample comes in, because a result you cannot attach to a person is a result you will quietly re-run.

PLAN CARD 119W

Day one, and the freight yard fire has been out for six hours. A row of containers 80 metres from the river is damaged and nobody can tell you what was in them. Three names are going around: one from a driver, one from a half-burnt manifest, one from a label nobody can read. Adaeze Okonjo, who leads the analytical chemistry, wants two methods on the same sample before anybody writes a name down; the fire chief wants a name now. Today you turn that into a short list you would defend, and decide how to spend the one round of lab work you get. Sampling has an order too, because some steps destroy what the next one needed.

BRIEFING 41W

A freight-yard fire damages several unlabeled containers beside Riverton's river. The response team must establish chemical identity and hazard classes before firefighters, residents, and the water utility act on rumors. A wrong identity can cause an incompatible firefighting or treatment decision.

WHAT YOU DECIDE 11W

Create a provisional identity list and choose the next discriminating measurements.

1.1 Read the formula, not the rumor

MOLECULAR IDENTIFICATION • MOLECULAR IDENTIFICATION LAB • A ROOM • PROTOCOL

SCENE 38W

A freight-yard fire has damaged several unlabelled containers, 80 metres from the river. Three different guesses about what is in them are already on the radio. Firefighters are picking a suppression agent from whatever identity you give them.

GUIDE 60W

The left column is four kinds of record. The right column is four kinds of claim. Ask of each record what it actually fixes: which elements, in what ratio, carrying what charge. A trade name fixes none of the three. Some substances on the shortlist react with water. A firefighter is choosing a suppression agent from whatever identity comes back.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a chemical formula lists which elements are present and in what ratio • atomic structure, ions and electron configuration — taken as read • periodic trends and what an element can do — taken as read

VERDICT 86W

A formula is not a name. It states which elements are present, in what whole-number ratio, and what charge the unit carries — and each of those constrains what the substance can do. Change a subscript or a charge and you have a different compound with different reactivity, different solubility and a different hazard class. A

common name carries none of that, which is why a label reading only a trade name cannot support a hazard prediction. Some of the wrong answers here react with water.

TAKEAWAY 8W

Chemical identity begins with explicit composition and charge.

1.2 From damaged container to provisional identity

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A PERSON · SEQUENCE

SCENE 49W

Inside the mobile laboratory, the first intact sample from the damaged drums is sealed on the bench. Its outside markings are legible now and will be scuffed off by the opening. Breaking the seal to draw liquid vents the headspace, and there is only enough liquid for two runs.

GUIDE 69W

Every one of these four is available right now, so nothing here is a clock. What separates them is what you can still do afterwards. Ask of each step what it consumes: the markings, the headspace, the liquid, or the sample itself. There is enough liquid for two runs. A step that ends the sample answers best and answers once, which puts it somewhere very particular in the order.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

some measurements leave the sample as they found it and some use it up

VERDICT 85W

The order is set by what each step costs. The markings on the outside of the drum survive only until somebody opens it, so they are recorded first and consume nothing. Headspace vapour samples what has already escaped, without breaching the bulk liquid. A non-destructive spectrum on a small aliquot can be repeated all week if it goes wrong. The destructive method gives the best identification and gives it once. It is therefore the last thing you spend and the first thing you would regret.

TAKEAWAY 12W

Analytical work runs from what costs nothing to what cannot be undone.

1.3 Spend the first of the lab budget

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A PERSON · VALUE

SCENE 42W

An \$800 laboratory budget is left for the first round of evidence. Three damaged drums have lost their labels, and the first extract is ready. Firefighters are waiting at the fence line for the provisional identity that will determine their next move.

GUIDE 62W

Open each evidence card to see what it would actually tell you, then spend up to \$800. A card's price is not its value. Firefighters at the fence need an identity firm enough to choose their next move, so buy the evidence that could distinguish between the candidates. Repeating a test you have already run buys certainty about something you already knew.

ASK 11W

Which evidence do you buy before giving firefighters a chemical identity?

BEHIND THE BUTTON 136W

What makes a test worth its money. Not precision, and not how much data it returns. The question is whether its two possible results point at different chemicals. A test every candidate would pass tells you nothing you did not know before you paid for it.

Why a provisional identity is enough. The firefighters do not need a publication-grade answer; they need to know which class of chemistry they are standing next to. Waiting for certainty means they act on nothing at all, which is the more dangerous of the two errors.

Why the labels being gone matters so much. Three drums, no labels, and one extract means the identity has to come out of the chemistry rather than the paperwork. Every card on the board is a way of asking the material what it is.

TAKES AS READ

different analytical methods can fail for different reasons · a repeated measurement is not automatically an independent measurement

VERDICT 78W

An analytical result can be precise and still be wrong. A repeat run tests repeatability, records test provenance, and quality controls test specific laboratory failure modes. None of those independently identifies the compound. A structural method based on different physics can overturn a mistaken retention-time match, so it changes the decision directly. With a limited budget, the first purchase should attack the uncertainty that can reverse the hazard call. Making the existing result look more complete is secondary.

TAKEAWAY 20W

Independent evidence is more valuable than a repeated result when the decision depends on whether the identity itself is right.

1.4 Water first, or never water

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A PERSON · BELT

SCENE 36W

The contractor lifts at four and wants the yard sorted by then. Some of these drums are made safe by dilution. Others react with water, and a hose is the worst thing that could reach them.

GUIDE 46W

Two bins. Something that simply dissolves is diluted by water and made less dangerous by it. Something that reacts with water makes heat, gas or both, so water is what turns a spill into an event. Sort on what the substance does when water reaches it.

ASK 14W

Send each drum to the bin that says what water would do to it.

BEHIND THE BUTTON 91W

Why this is the first question at a yard. The default response to almost anything is water — hoses are what a fire crew has. For most of what is here that is right, and for a short list it is the one action that makes the situation worse.

What reacting with water looks like. Alkali metals and hydrides give off hydrogen. Concentrated acids give off enough heat to boil what is poured onto them. Carbides and some chlorides give off gas that is worse than what was in the drum.

TAKES AS READ

some substances dissolve in water and some react with it · atomic structure, ions and electron configuration — taken as read

VERDICT 166W

Water is the default response to a spill because it is what everybody has, and for most of a yard it is the right one: a soluble salt or an acid already in dilute solution is made less dangerous by more water, and the concentration is the hazard. The short list is where the default inverts. Alkali metals and hydrides displace hydrogen from water and give off enough heat to light it. Concentrated sulphuric acid releases so much heat on dilution that water poured onto it boils and throws the acid back — which is why the rule is acid into water and never the reverse. Carbides give off acetylene, and some chlorides give off hydrogen chloride gas that is worse in air than the drum was on the ground. Nothing about the drums looks different. The label is the only thing that tells you which list a drum is on, which is why a yard of unlabelled drums is the emergency rather than an inconvenience.

TAKEAWAY 14W

The default response is right for most things and catastrophic for a short list.

END OF THE DAY 11W

Create a provisional identity list and choose the next discriminating measurements.

The Vapor Cloud

BEFORE THE DAY — TRIAL

Walk Riverton before you sample it

The city is the reason this is hard: a river through the middle, a works upstream of a reservoir, and a container yard nobody has inventoried. Walk the sites once now, while nothing is urgent, because a sampling plan drawn from a map puts points where the roads are rather than where the water goes.

PLAN CARD 111W

Day 2, first light, and the wind has backed 40 degrees. A colourless plume is drifting off the yard toward 2 neighbourhoods, and the forecast from the atmospheric chemistry lead, Miklos Varga, disagrees with the incident commander's map about where it goes by 11. Sipho Mbeki, who briefs the public, has to tell residents something by eight. Today you work out how much air that release can fill, what the warming morning will do to it, and which measurement to buy next. Both mistakes cost people. Evacuate too small an area and somebody is standing downwind of it. Evacuate the whole district and you block the roads the fire crews need.

BRIEFING 44W

A colorless plume moves toward two neighborhoods as the weather changes. The team must estimate expansion, density, and direction without treating an ideal-gas model as a complete atmospheric forecast. A delayed evacuation risks exposure; an overbroad evacuation can block emergency access and overwhelm shelters.

WHAT YOU DECIDE 11W

Provide a bounded plume assessment and decision triggers for evacuation zones.

2.1 How much volume can the gas occupy?

ATMOSPHERE & GASES · AIR & PLUME STATION · A ROOM · BALLPARK

SCENE 41W

At the command trailer, a colourless plume is moving toward two neighbourhoods. Police are holding the evacuation roads open while the wind shifts. The release inventory is finally known, and the incident commander wants the gas scale before moving the roadblocks.

GUIDE 62W

Five numbers, and two of them describe conditions the release is not at. One is the molar volume at standard temperature and pressure. One is zero Celsius in kelvin. The relationship is written for ambient conditions, so every term has to be the ambient one. A volume computed at the wrong conditions still looks like a volume. Roadblocks get moved on it.

ASK 7W

Estimate the gas volume at ambient conditions.

BEHIND THE BUTTON 134W

What the relationship says. $V = nRT/P$, with V the volume, n the moles of gas, R the gas constant, T the absolute temperature and P the pressure.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a given amount of gas occupies a predictable volume at ordinary conditions

VERDICT 88W

The ideal gas law turns an amount of gas into a volume you can reason about. At ordinary temperature and pressure, the scale is about 24 litres per mole. It is a first-order tool and nothing more: it says how big the release is, not where it goes. The hazardous footprint depends on mixing, wind, terrain and chemistry, none of which appear in the calculation. The number still has to be right, because evacuating too little exposes people and evacuating too much blocks the roads the response needs.

TAKEAWAY 10W

Simple gas laws provide scale while atmospheric transport determines exposure.

2.2 Interpret plume behavior

ATMOSPHERE & GASES · AIR & PLUME STATION · A PERSON · PROTOCOL

SCENE 38W

The weather station shows rising temperature, falling pressure and a strengthening afternoon wind while the plume is still moving. Ground crews also report vapour pooling in a low drainage channel on the downwind side of the freight yard.

GUIDE 69W

Two different physics run at once here, and the four situations do not all belong to one. Some change what a fixed amount of gas occupies. Others change where the parcel goes. Ask which of the two each one acts on before pairing it. Read a transport change as a gas-law change and the corridor is drawn from the wrong physics. The vapour is pooling in a drainage channel.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a gas has a state and also a place, and they change for different reasons

VERDICT 85W

Two different physics are running at once. Gas-law behaviour says how a fixed amount of substance responds to temperature and pressure. Charles and Boyle are two statements about the same gas, one proportional and one inverse. Atmospheric transport says where the parcel actually goes. Density relative to air affects whether the cloud hugs the ground or lifts. Wind and turbulence control dilution, while terrain controls where vapour can pool. Each change on the board pushes on one part of that system, not necessarily the others.

TAKEAWAY 9W

A plume is a coupled thermodynamic and transport problem.

2.3 Why the drum is worse in the afternoon

ATMOSPHERE & GASES · AIR & PLUME STATION · A PERSON · CHOICE

SCENE 35W

The same open drum reads three times the vapour concentration at two in the afternoon as it did at six in the morning. Nothing has been added to it and nothing has been taken out.

GUIDE 72W

All four options would explain a rising reading, and they differ in what has changed. One changes the liquid, one changes the instrument, one changes the amount, and one changes only the partition between liquid and air. Nothing was added and nothing was removed, which is the fact that does the work. Getting this wrong sets an exposure limit from a dawn reading and clears a site that is worse by mid-afternoon.

ASK 9W

Same drum, same liquid, three times the vapour. Why?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a liquid and its vapour sit in equilibrium at a temperature

VERDICT 88W

Vapour pressure is the pressure of gas in equilibrium with its own liquid, and it climbs steeply with temperature — often doubling for every 10 or 15 degrees.

Warming the drum from morning to afternoon moves more molecules into the gas phase above the same liquid, so the air above it is more concentrated with no change in what is in the drum. This is why exposure limits are checked at the hottest part of the day, and why a hazard assessment made at dawn understates the afternoon.

TAKEAWAY 18W

How much of a liquid is in the air above it depends steeply on how warm it is.

END OF THE DAY 11W

Provide a bounded plume assessment and decision triggers for evacuation zones.

A Reaction Underground

BEFORE THE DAY — FOLLOW

Ferreira walks the river reach once

Where the plume actually enters the river is not on any drawing at the accuracy anybody needs, and Ferreira, the water and sediment lead, can read it off the bank — the colour of the silt, where the weed stops. She walks it once this morning. Stay with her, close enough to see what she is pointing at and far enough back not to be standing in it.

PLAN CARD 96W

Day 3. Two drains may be meeting in a closed tunnel under the yard and the closed-space safety officer, Femi Oyelaran, has a crew suited up at the hatch waiting on your word. He will not send them in on a story. Oyelaran has monitors at the hatch and two hours of bottled air per man. Today you work out the largest reaction that could be happening down there, and whether the reaction you assumed is the one really going on. What hurts people in a closed space is ordinary: heat, pressure, and oxygen pushed out.

BRIEFING 39W

Two drainage streams may mix in a confined tunnel. The player must determine limiting reactants, plausible products, and gas or heat generation before crews enter. Entering before bounding the chemistry could expose crews to heat, pressure, or oxygen displacement.

WHAT YOU DECIDE 16W

Estimate the maximum reaction scale and define measurements that verify whether the assumed chemistry is occurring.

3.1 From identity to reaction estimate

REACTIONS & ENERGY · REACTION & THERMAL HALL · A ROOM · SEQUENCE

SCENE 41W

At the tunnel entrance, two drainage streams may be meeting behind a closed access door. A repair crew is waiting at the barricade, and the storm bypass remains shut. The laboratory has analyses from both drains but no reaction estimate yet.

GUIDE 66W

These four are not four independent tasks. Each needs the output of another. Ask of each card what it requires before it can be done at all. Coefficients cannot be used until the equation is balanced. Masses cannot be compared until they are moles. A prediction is worth making only once there is a measurement to compare it against. A repair crew is at the barricade.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a balanced equation says how many of one thing react with how many of another

VERDICT 74W

Every quantitative reaction prediction begins with a balanced equation. Its coefficients supply the mole ratios between reactants and products. Measured masses are then converted to moles, because the coefficients compare particles rather than grams. Those mole ratios identify which reactant runs out first. Only after the limiting reactant is known can the maximum products be calculated. The prediction can then be checked against the gas, heat or remaining reagent actually observed in the tunnel.

TAKEAWAY 10W

Stoichiometry is a conditional prediction based on a specified reaction.

3.2 Which reactant limits?

REACTIONS & ENERGY · REACTION & THERMAL HALL · A PERSON · BALLPARK

SCENE 32W

A reaction takes two moles of A for every mole of B, and both are arriving in the tunnel in amounts nobody chose. Closed-space entry decisions are being made from this number.

GUIDE 65W

Four numbers, and one of them is the two reactant amounts added together. Another is the reaction's own ratio rather than a quantity of anything. A balanced equation compares particles, so amounts have to be in moles before the ratio can be applied. Adding two reactants gives a number that no stoichiometry uses, and closed-space entry decisions are being made from whichever number comes out.

ASK 14W

Which reactant limits, how much B can react, and what percent yield was recovered?

BEHIND THE BUTTON 133W

What the relationship says. First use $n = m/M$. Then compare mole amounts with the 2 A : 1 B coefficients. Percent yield = actual/theoretical $\times 100$.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a reaction stops when one ingredient runs out

VERDICT 72W

A balanced equation compares particles, so masses have to become moles before the reactants can be compared. 500 grams of A at 50 g/mol is 10 mol. 640 grams of B at 80 g/mol is 8 mol. The 2:1 ratio means 10 mol A can consume only 5 mol B, so A limits. If the recovered product corresponds to 4.2 mol of reaction, the percent yield is $4.2/5.0 \times 100 = 84\%$.

TAKEAWAY 17W

Stoichiometric limits are set in moles, and percent yield compares what was recovered with that theoretical limit.

3.3 Heat release is not the spontaneity test

REACTIONS & ENERGY • REACTION & THERMAL HALL • A PERSON • BALLPARK

SCENE 43W

A clean-up reagent is staged beside the tunnel, but the tank test has shown no product after 6 days at 18 °C. Its measured ΔH and ΔS are on the clipboard. Operations wants to know whether the chemistry is impossible or simply slow.

GUIDE 69W

Four numbers, and two of them are the same temperature written in different scales. The relationship needs an absolute temperature, so degrees Celsius put straight into it is not a small error but a different quantity. Note what the answer does and does not settle: a favourable sign says the reaction can go, not that it will go this week. The tank has had six days and shows nothing.

ASK 18W

At 18 °C, what is ΔG , and what does the lack of product say about the reaction rate?

BEHIND THE BUTTON 119W

What the relationship says. $\Delta G = \Delta H - T\Delta S$, with T in kelvin.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

temperature in $\Delta G = \Delta H - T\Delta S$ must be in kelvin • a negative ΔG means the forward reaction is thermodynamically favored

VERDICT 70W

Free energy combines enthalpy, entropy and temperature. At 291 K, $\Delta G = -180 - 291(-0.40) \approx -64$ kJ/mol, so the reaction is thermodynamically favorable. That does not make it fast. The unchanged tank points to a substantial activation barrier at this temperature. This distinction matters in treatment design. Temperature or a catalyst can change reaction rate without changing the sign of ΔG . A favorable ΔG alone never guarantees useful speed.

TAKEAWAY 14W

Free energy sets thermodynamic favorability; activation barriers set how quickly a favorable reaction proceeds.

3.4 Read the formula, not the rumor

MOLECULAR IDENTIFICATION • MOLECULAR IDENTIFICATION LAB • A ROOM •
CALLBACK • PROTOCOL

SCENE 38W

A freight-yard fire has damaged several unlabelled containers, 80 metres from the river. Three different guesses about what is in them are already on the radio. Firefighters are picking a suppression agent from whatever identity you give them.

GUIDE 60W

The left column is four kinds of record. The right column is four kinds of claim. Ask of each record what it actually fixes: which elements, in what ratio, carrying what charge. A trade name fixes none of the three. Some substances on the shortlist react with water. A firefighter is choosing a suppression agent from whatever identity comes back.

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TAKES AS READ

a chemical formula lists which elements are present and in what ratio • atomic structure, ions and electron configuration — taken as read • periodic trends and what an element can do — taken as read

VERDICT 86W

A formula is not a name. It states which elements are present, in what whole-number ratio, and what charge the unit carries — and each of those constrains what the substance can do. Change a subscript or a charge and you have a different compound with different reactivity, different solubility and a different hazard class. A

common name carries none of that, which is why a label reading only a trade name cannot support a hazard prediction. Some of the wrong answers here react with water.

TAKEAWAY 8W

Chemical identity begins with explicit composition and charge.

END OF THE DAY 16W

Estimate the maximum reaction scale and define measurements that verify whether the assumed chemistry is occurring.

Identify the Unknowns

PLAN CARD 100W

Day 4. The lab has a chromatogram — a chart with one peak for each thing it managed to separate out — and there are more peaks than the team has names. Two methods run on the same sample disagree. Kwabena Osei, the treatment engineer, wants a list this morning so the plant can start ordering chemistry; Okonjo will not give him one she cannot defend. Today you decide what is actually in the mixture and how sure you are willing to be about each name. Name something that is not there and the plant treats the water for it.

BRIEFING 44W

Chromatography reveals several peaks, while spectroscopy suggests overlapping functional groups. The player must combine separation and structural evidence rather than forcing one peak to equal one compound. A false identification could lead the treatment plant to use chemistry that creates a more hazardous byproduct.

WHAT YOU DECIDE 9W

Assign confidence-ranked identities to mixture components using independent evidence.

4.1 What the position in the table tells you

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM · CHOICE

SCENE 34W

Two unlabelled drums, one holding a sodium salt and one a caesium salt of the same anion. The bench has to say which will be the more violent with water before either is opened.

GUIDE 66W

Two drums share an anion, so the anion cannot separate them. Each option offers a different basis for the ranking: position in a group, atomic mass, the shared part, or refusal to predict. Ask what changes down group 1, and why. The table exists to rank two substances a bench has never handled. That ranking is what a crew works to before either drum is opened.

ASK 12W

Which drum is the more reactive with water, and on what grounds?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

the periodic table groups elements that behave alike · atomic structure, ions and electron configuration — taken as read

VERDICT 79W

Group 1 reactivity rises going down: the outer electron sits further from the nucleus and is more shielded, so it is lost more easily. Caesium is below sodium, so the caesium salt is the more energetic with water. The anion is the same in both, which is exactly why the cation decides it. This is what the table is for — it lets a bench rank two substances it has never handled, from position alone, before anything is opened.

TAKEAWAY 11W

An element's position in the periodic table predicts how it behaves.

4.2 Read analytical disagreement

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A PERSON · TRACE

SCENE 37W

Two methods have been run on the same extract and they do not tell the same story. A false identification here sends the plant a chemistry that could create a worse product than the one it removes.

GUIDE 50W

Open each channel to see what its result depended on. Keep the ones that stand on an independent measurement, and untick the ones whose chain runs through a shared step. Then name the step they have in common. Two methods disagreeing is information, not an error to be averaged away.

ASK 11W

Which channels remain trustworthy, and what shared source is at fault?

BEHIND THE BUTTON 143W

Why disagreement is the useful signal. Two methods agreeing can mean both are right or both inherit the same mistake. Two disagreeing means at least one assumption is wrong, and finding which is a question about what each method depended on rather than about which number looks better.

What a shared step looks like here. A calibration standard, a dilution, a column, an internal reference: anything both methods passed through. Everything downstream of it moves together if it is wrong, which is why the average of two such results is no safer than either.

What is riding on getting it right. A false identification sends the plant a treatment chemistry chosen for the wrong compound, and the product of that reaction can be worse than what it removes. Withdrawing both results is safer than averaging them and worse than naming the shared step.

TAKES AS READ

a blank contains every preparation step except the sample · independent methods do not share the same critical failure path

VERDICT 80W

The sample and duplicate chromatograms agree because they share the same preparation solvent, retention reference and detector path. Their agreement therefore does not provide two independent identifications. The blank carries almost the same 3.1-minute feature even though it contains no river sample, which points to preparation contamination. The structural spectrum uses a different measurement path and does not match the target. Together, the blank and the independent spectrum overturn the retention-time identification while leaving the rest of the run interpretable.

TAKEAWAY 16W

Agreement counts as independent evidence only when the measurements do not inherit the same failure path.

4.3 Resolve the ambiguous peak

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A PERSON · CHOICE

SCENE 39W

1 peak at 4.2 minutes matches 2 candidates to within 0.05 minutes, and both are on the shortlist. The river sample also carries humic matter, chloride at 40 mg/L, and a detergent from the foam used on the fire.

GUIDE 62W

Two candidates come off the column together. So anything that depends on how fast they travel inherits the ambiguity. Sort the options by that test. Does this one improve the separation, replace it with evidence of another kind, add controls, or only change the presentation? Controls make a number defensible and cannot resolve an overlap. Better presentation adds no measurement at all.

ASK 15W

One peak is still ambiguous and the review board wants it settled. What settles it?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

two compounds that overlap in one method may not overlap in another

VERDICT 84W

There are two honest ways out of an overlap. Improve the separation until the components stop coming off the column together, or bring in evidence of a different kind so the answer no longer depends on the separation at all. The second is faster, and it cannot inherit the problem: a method that identifies by structure does not care that two compounds travelled at the same speed. The third way is to present the same weak data more attractively, and it adds no measurement.

TAKEAWAY 15W

Better presentation cannot repair a method that was fooled, or evidence that was never validated.

4.4 How much volume can the gas occupy?

ATMOSPHERE & GASES · AIR & PLUME STATION · A ROOM · CALLBACK · BALLPARK

SCENE 41W

At the command trailer, a colourless plume is moving toward two neighbourhoods. Police are holding the evacuation roads open while the wind shifts. The release inventory is finally known, and the incident commander wants the gas scale before moving the roadblocks.

GUIDE 62W

Five numbers, and two of them describe conditions the release is not at. One is the molar volume at standard temperature and pressure. One is zero Celsius in kelvin. The relationship is written for ambient conditions, so every term has to be the ambient one. A volume computed at the wrong conditions still looks like a volume. Roadblocks get moved on it.

ASK 7W

Estimate the gas volume at ambient conditions.

BEHIND THE BUTTON 134W

What the relationship says. $V = nRT/P$, with V the volume, n the moles of gas, R the gas constant, T the absolute temperature and P the pressure.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a given amount of gas occupies a predictable volume at ordinary conditions

VERDICT 88W

The ideal gas law turns an amount of gas into a volume you can reason about. At ordinary temperature and pressure, the scale is about 24 litres per mole. It is a first-order tool and nothing more: it says how big the release is, not where it goes. The hazardous footprint depends on mixing, wind, terrain and chemistry, none of which appear in the calculation. The number still has to be right, because evacuating too little exposes people and evacuating too much blocks the roads the response needs.

TAKEAWAY 10W

Simple gas laws provide scale while atmospheric transport determines exposure.

END OF THE DAY 9W

Assign confidence-ranked identities to mixture components using independent evidence.

What Dissolved in the River?

PLAN CARD 103W

Day 6. Runoff reached the river on the first night and Riverton's drinking-water intake is 900 metres downstream. Inês Ferreira, who owns water and sediment, has been out since four and wants it sampled; the state's auditor wants a water survey, because that is what a water survey means. Today you design it: where to sample, in what order, and which parts of the river to sample at all. That last one decides everything else. Test only the water and the result comes back clean, and it is honestly clean — today. The next flood will stir that sediment back into the water.

BRIEFING 52W

Runoff from the accident enters the river. Some compounds remain in surface films, some dissolve, and others bind to sediment. The player must predict partitioning from molecular interactions and verify it with sampling. If the team samples only water, a persistent sediment reservoir may be missed until it re-enters the water supply.

WHAT YOU DECIDE 14W

Predict where each chemical class will be found and design a mass-balance sampling plan.

5.1 Where will the chemical go?

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A ROOM · PROTOCOL

SCENE 34W

Runoff from the accident has reached the river. If you predict the wrong phase you will sample the wrong thing, and a clean water result will be read by the city as an all-clear.

GUIDE 69W

The left column is four molecular structures and the right column is four places a chemical can end up. Pair them by asking what water can do for that structure: hydrate an ion, hydrogen-bond a polar molecule, or refuse a non-polar one. A compound that avoids water sits on the particles instead. Sample the wrong phase and a clean water result gets read by the city as an all-clear.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

like dissolves like — polar mixes with polar, and non-polar does not · atomic structure, ions and electron configuration — taken as read

VERDICT 81W

A chemical released into water does not stay in one place or one phase, and where it goes follows from molecular structure. Polar molecules that can hydrogen-bond dissolve readily. Ionic species dissolve as hydrated ions, surrounded by water molecules. Non-polar liquids, which barely mix with water, separate out or spread as a surface film. Compounds that avoid water partition onto organic carbon in

suspended particles and sediment and stay there. Absence from the water column is not absence from the river.

TAKEAWAY 13W

The absence of a chemical from water does not mean it has disappeared.

5.2 Use the common-ion effect to clear the limit

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A PERSON · CHOICE

SCENE 38W

The filtered water has 3.1 mg/L of dissolved metal against a 0.5 mg/L permit limit. The solid is a 1:1 metal carbonate and pH is fixed. The pilot can raise dissolved carbonate eightfold without changing the water volume.

GUIDE 65W

All four options give a number and a reason. The reasons are what differ: fixed solubility, a product held constant, both ions rising, or a squared dependence. Check the stoichiometry of the solid first. The exponents in the solubility product come from it. This has to clear a permit limit. An answer out by a factor of eight is a discharge that fails its consent.

ASK 10W

Eight times the carbonate: what happens to the dissolved metal?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a saturated solution is in equilibrium with a 1:1 solid · K_{sp} stays constant when temperature stays fixed

VERDICT 72W

For a 1:1 metal carbonate, the solubility product is proportional to the dissolved metal concentration times the carbonate concentration. At fixed temperature, that product stays constant at equilibrium. Raising carbonate by a factor of eight therefore drives the dissolved metal down by the same factor. Starting from 3.1 mg/L gives about 0.39 mg/L. That is below the 0.5 mg/L limit, although a real pilot still has to check pH, complexation and settling.

TAKEAWAY 19W

For a 1:1 sparingly soluble salt, increasing 1 ion lowers the equilibrium concentration of the other in inverse proportion.

5.3 Spend the river survey budget

SOLUTIONS & WATER CHEMISTRY • WATER INTAKE LABORATORY • A PERSON •
ALLOCATE

SCENE 37W

The survey has 20 samples in the budget. The compound is sparingly soluble, and the river runs 900 metres from the release to the intake. Two grab samples taken so far disagree by a factor of six.

GUIDE 58W

Twenty bottles is the whole survey. Each package you select spends some of them, and the panel shows which river questions your current plan can answer. Watch that list rather than the map. Two grab samples already disagree by a factor of six, so a plan that cannot explain why is a plan that will be argued with.

ASK 13W

How do you spend 20 bottles without leaving a required river question unanswered?

BEHIND THE BUTTON 148W

Why the disagreement is the design brief. A factor of six between two grabs means the compound is not evenly mixed. That can be plume structure across the channel, depth stratification, or a change with time. Each of those is a different sampling pattern, and one grab per site cannot tell them apart.

Why sparing solubility matters. A compound that barely dissolves travels partly as droplets or on particles rather than in solution, so a filtered bottle and an unfiltered one answer different questions. Nine hundred metres is far enough for that difference to matter at the intake.

What the required questions are for. Not everything the survey could ask, but the ones a decision hangs on: whether the intake is affected now, whether it will be, and by how much. A plan that answers other things beautifully and misses one of those has spent twenty bottles badly.

TAKES AS READ

different sample locations and phases answer different questions • a clean water sample cannot rule out contaminant stored in solids

VERDICT 83W

Environmental sampling needs coverage across the dimensions that can change the conclusion. Intake samples measure current exposure, upstream samples establish background, and solids plus sediment test whether a water-avoiding compound has left the water column. A visually stained bank may be interesting, but it is not required for the main decision. The survey must test movement toward the intake and storage in another phase. The twenty-bottle limit forces those priorities to be explicit instead of letting every desirable sample appear on the plan.

TAKEAWAY 22W

A sampling plan is a set of questions it can answer, and every bottle spent on one question is unavailable to another.

5.4 From damaged container to provisional identity

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM · CALLBACK · SEQUENCE

SCENE 49W

Inside the mobile laboratory, the first intact sample from the damaged drums is sealed on the bench. Its outside markings are legible now and will be scuffed off by the opening. Breaking the seal to draw liquid vents the headspace, and there is only enough liquid for two runs.

GUIDE 69W

Every one of these four is available right now, so nothing here is a clock. What separates them is what you can still do afterwards. Ask of each step what it consumes: the markings, the headspace, the liquid, or the sample itself. There is enough liquid for two runs. A step that ends the sample answers best and answers once, which puts it somewhere very particular in the order.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

some measurements leave the sample as they found it and some use it up

VERDICT 85W

The order is set by what each step costs. The markings on the outside of the drum survive only until somebody opens it, so they are recorded first and consume nothing. Headspace vapour samples what has already escaped, without breaching the bulk liquid. A non-destructive spectrum on a small aliquot can be repeated all week if it goes wrong. The destructive method gives the best identification and gives it once. It is therefore the last thing you spend and the first thing you would regret.

TAKEAWAY 12W

Analytical work runs from what costs nothing to what cannot be undone.

END OF THE DAY 14W

Predict where each chemical class will be found and design a mass-balance sampling plan.

The Concentration Problem

BEFORE THE DAY — HUNT

Six drums on the manifest, four in the yard

The manifest says six drums came off the lorry and the yard holds four. Until the other two are found nobody can say what is in the ground, because a drum nobody has located is a drum that is either in a skip or leaking somewhere nobody is sampling. Find them.

PLAN CARD 119W

Day 8, and this is where the utility director, Marisol Reyes, and her quality assurance lead, Rie Nakamura, stop agreeing. The results are back as numbers and the city is holding each one against a legal limit, one water system at a time. Nakamura wants every result re-run before anything reopens. Reyes points out that four systems are shut on numbers nobody has questioned, hospitals are on tankers, and a week of caution has a cost too — and today she is right. Today you turn readings into what was really in the river, decide whether one result can be trusted, and choose where to sample. Outside the calibrated range nobody has checked what the instrument does at all.

BRIEFING 48W

City leaders ask whether the contaminant is "high," but samples differ in dilution, path length, and what else is in the sample. The team must turn signals into defensible concentrations. An incorrect concentration map can either close a safe water system or expose residents through a false reassurance.

WHAT YOU DECIDE 12W

Produce concentration maps whose units, detection limits, and quality controls are explicit.

6.1 Undo the dilution

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A ROOM · BALLPARK

SCENE 32W

The instrument reports what was in the vial, not what was in the river. City leaders are comparing your number against a threshold and closing or opening a water system with it.

GUIDE 67W

Five numbers, and two of them belong to different questions: a detection limit and the volume of the bottle. Two others are the volumes before and after the dilution, and only their ratio matters. Ask of each number whether it describes the vial, the river, or the method. A dilution not carried back understates the river tenfold, and nothing about the reported number says it was diluted.

ASK 4W

Estimate the original concentration.

BEHIND THE BUTTON 144W

What the relationship says. The concentration of the solution is the amount dissolved divided by the volume holding it, so $\text{amount} = C \times V$. Diluting adds solvent and not solute, which leaves $C_{\text{original}} V_{\text{sample}} = C_{\text{diluted}} V_{\text{final}}$.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

diluting a sample lowers its concentration by a known factor

VERDICT 74W

Samples are diluted so the signal lands inside the range the method can measure. That factor has to be carried back through before the number says anything about the river. A tenfold dilution reported as measured understates the river ten times over — and the report still looks like a valid result, because nothing about the number says it was diluted. Preparation is part of the answer, not something that happened to it first.

TAKEAWAY 8W

Sample preparation is part of the quantitative result.

6.2 Protect the calibration

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A PERSON · VERIFY

SCENE 32W

An instrument will return a number for almost anything you put in front of it. The panel in front of you has everything needed to decide whether this reading can be defended.

GUIDE 55W

Lock the absorbance you expect after a one-to-one dilution before you make it. Then make the dilution and read what the instrument returns. The last step costs a verification slot, and spending it is the point. An instrument returns a number for almost anything, and only the comparison says whether this one can be defended.

ASK 15W

What should the absorbance become after dilution, and does the rerun support the dilution model?

TAKES AS READ

Beer–Lambert gives A proportional to concentration when path length and absorptivity are fixed · a 1:1 dilution halves concentration

VERDICT 77W

Beer–Lambert predicts absorbance proportional to concentration while the method remains linear. A 1:1 dilution halves concentration, so an absorbance of 1.34 should fall to about 0.67. The observed 0.68 is inside the validated calibration range and agrees with that prediction. The diluted concentration can then be read from the calibration and doubled for the original sample. This is stronger than extrapolating 1.34 because both the dilution and the response are checked inside the range established by standards.

TAKEAWAY 22W

A measurement outside its calibrated range becomes defensible only after a controlled change moves it into-range and the predicted response is verified.

6.3 Map concentration with limited samples

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A PERSON · CHOICE

SCENE 41W

The city wants a concentration map of 11 kilometres of river, and the laboratory can run 30 samples. The 2 known hot spots are 400 metres apart, and the field kit reads to plus or minus 30% against the laboratory's 5.

GUIDE 65W

Every option here buys something real, and they trade against one another. Coverage of the river runs against confidence in each number, and both against thirty samples. Ask of each what decision it improves. Perfect chemistry where nobody drinks misleads. So does dense coverage with an unproven method. Every decision the city makes happens in one place, and detail elsewhere is detail nobody acts on.

ASK 12W

You cannot sample every metre of river. Where do the samples go?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a map is only worth what the decisions it supports are worth

VERDICT 76W

A map trades off two things that both cost money. Analytical validity is whether each number can be trusted. Spatial coverage is whether the numbers sit where the decision is made. Neither alone protects anybody: perfect chemistry at the wrong points and dense sampling with an unproven method both produce a confident map that misleads. Every decision the city is about to make happens at the intake, so detail anywhere else is detail nobody acts on.

TAKEAWAY 12W

Spatial coverage without analytical validity—or validity without spatial coverage—cannot protect the city.

6.4 What the bench is running now

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A PERSON · SPOT

SCENE 35W

Samples queue on the bench all day. The laboratory manager holds one priority, and changes it as the incident does — after the river result, after the schools ask, after the hold times start expiring.

GUIDE 53W

Run the samples the current priority wants and leave the rest queued. The priority on the bench card is changed during the day and nobody announces it. What is scored is the samples either side of a change, because they are the only ones that show whether the bench is reading the card.

ASK 13W

Run to the priority on the bench card, and keep watching the card.

BEHIND THE BUTTON 91W

Why a laboratory priority changes. Early it is whatever the incident commander is waiting on. Then it is whatever is closest to a hold time, because a sample analysed after its hold time is not a result. Then it is whatever a decision is actually waiting on, which is rarely the same list.

Why the changeover costs. A bench working to the old priority for another twenty minutes spends its one clean instrument on a sample nobody is waiting for, and the sample that was waiting expires quietly in the fridge.

TAKES AS READ

a laboratory works to a stated priority that somebody can change

VERDICT 141W

Three priorities run across the day and each wants a different part of the rack. Anything the commander is waiting on, then anything near its hold time, then anything from a drinking water source. A sample can answer two at once, which is what makes the change cost real. The panel scores the window either side of each change, because most of the rack is wanted by neither priority and is correctly left queued by somebody who has read nothing. The laboratory's own version is the twenty minutes after the schools ask: the priority has changed to drinking water, and the instrument is still working through river samples that were urgent an hour ago. And the cost is invisible afterwards — a sample past its hold time comes out of the fridge looking exactly like a sample that is still valid.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

6.5 Spend the first of the lab budget

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM · CALLBACK · VALUE

SCENE 42W

An \$800 laboratory budget is left for the first round of evidence. Three damaged drums have lost their labels, and the first extract is ready. Firefighters are waiting at the fence line for the provisional identity that will determine their next move.

GUIDE 62W

Open each evidence card to see what it would actually tell you, then spend up to \$800. A card's price is not its value. Firefighters at the fence need an identity firm enough to choose their next move, so buy the evidence that could distinguish between the candidates. Repeating a test you have already run buys certainty about something you already knew.

ASK 11W

Which evidence do you buy before giving firefighters a chemical identity?

BEHIND THE BUTTON 136W

What makes a test worth its money. Not precision, and not how much data it returns. The question is whether its two possible results point at different chemicals. A test every candidate would pass tells you nothing you did not know before you paid for it.

Why a provisional identity is enough. The firefighters do not need a publication-grade answer; they need to know which class of chemistry they are standing next to. Waiting for certainty means they act on nothing at all, which is the more dangerous of the two errors.

Why the labels being gone matters so much. Three drums, no labels, and one extract means the identity has to come out of the chemistry rather than the paperwork. Every card on the board is a way of asking the material what it is.

TAKES AS READ

different analytical methods can fail for different reasons · a repeated measurement is not automatically an independent measurement

VERDICT 78W

An analytical result can be precise and still be wrong. A repeat run tests repeatability, records test provenance, and quality controls test specific laboratory failure modes. None of those independently identifies the compound. A structural method based on different physics can overturn a mistaken retention-time match, so it changes the decision directly. With a limited budget, the first purchase should attack the uncertainty that can reverse the hazard call. Making the existing result look more complete is secondary.

TAKEAWAY 20W

Independent evidence is more valuable than a repeated result when the decision depends on whether the identity itself is right.

END OF THE DAY 12W

Produce concentration maps whose units, detection limits, and quality controls are explicit.

The Water Changes pH

PLAN CARD 104W

Day 11. The intake has turned acidic and Reyes wants a neutralising dose this morning; Novák, who maintains the pipeline, wants to know what it will do to her mains before anybody doses anything. The titration Ferreira ran overnight is nearly flat for four millilitres and then turns over hard. Today you work out how much acid is really there, read what that curve says about this water, and pick a dose that will not overshoot. There is one dosing pump and no way to take a dose back out. Overshoot and metals come off the pipe walls. Novák's question is the right one.

BRIEFING 47W

The river intake becomes acidic, while one proposed treatment could overshoot into strongly basic conditions. The player must use acid-base equilibria and buffering to control pH without masking chemical risk. A pH correction can mobilize metals, damage infrastructure, or injure workers if the chemical context is ignored.

WHAT YOU DECIDE 13W

Select a controlled neutralization strategy and determine when buffer capacity will be exhausted.

7.1 How many moles of acid are present?

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A ROOM · BALLPARK

SCENE 35W

The river intake is acidic and the plant is preparing a neutralisation dose. A one-thousand-litre test volume has a measured pH of 4.0. Operators want the size of the free hydrogen-ion pool before adding base.

GUIDE 65W

Four numbers, and two of them are the pH reading itself and the concentration in neutral water. A pH is a logarithm, so the reading cannot enter a mole calculation as a concentration. Convert first, then multiply by the volume. Note what the answer covers. This is the free hydrogen ion at this moment. A buffered water can release much more as base goes in.

ASK 11W

How many moles of free H^+ does the pH reading represent?

BEHIND THE BUTTON 121W

What the relationship says. $pH = -\log[H^+]$, so $[H^+] = 10^{(-pH)}$. Then $n = cV$.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

pH is a logarithmic scale — one unit is a factor of ten

VERDICT 73W

A pH of 4.0 means $[H^+] = 10^{-4}$ mol/L, not 4 mol/L and not 4 times neutral water. Multiplying that concentration by 1,000 L gives 0.10 mol of free H^+ . That number is useful but incomplete. Weak acids and buffers can release more H^+ as base is added, so pH does not measure the full neutralisation demand. A titration does, which is why the later dosing decision cannot come from this calculation alone.

TAKEAWAY 19W

pH is logarithmic, and converting it to moles still captures only the free hydrogen ion present at that moment.

7.2 Read the titration curve

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A PERSON · DIAGNOSIS

SCENE 37W

A trial dose calculated from the intake pH overshoot badly. The bench titration is now beside a strong-acid reference curve. Fresh pH 4 and pH 7 calibration checks from the same electrode are also on the screen.

GUIDE 74W

Six readings, and two of them exist to rule things out: the fresh calibration checks and the strong-acid reference at the same pH. Compare the shape of the trial curve against that reference before choosing. A long plateau and an equivalence point above neutral are both features a strong acid does not have. The trial dose overshoot, which means whatever explains the curve also has to explain a demand the starting pH never showed.

ASK 22W

The titration curve is on screen. Which explanation accounts for the plateau, the equivalence point and the dose the trial actually needed?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a weak acid holds most of its acidity undissociated

VERDICT 77W

A titration curve shows how much base the water consumes as pH changes. A long plateau means a buffer is absorbing added base while pH moves slowly. An equivalence point above pH 7 is also diagnostic of a weak acid system. The conjugate base left near equivalence is itself basic. A strong acid reference reaches equivalence

sharply and with much less added base. The curve therefore measures total neutralisation demand that the starting pH alone cannot reveal.

TAKEAWAY 10W

Titration is both a quantitative method and a diagnostic fingerprint.

7.3 Control pH without overshoot

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A PERSON · TRIGGER

SCENE 37W

The base pump is primed and operators are waiting to open the valve. Intake water is at pH 4.8, the cast-iron line is corroding, and every pump change appears 1 sampling interval later on the mixed-tank electrode.

GUIDE 86W

One rule, written before the run and fixed the moment you release it. The base feed is pushing the tank up, and anything past pH 7.5 breaches consent from the other side. The catch is delay: a pump command shows on the mixed-tank electrode one reading later, so the tank goes on climbing after the feed is off. Put the line back from 7.5 by that one reading of travel — and not so far back that it fires on water which has shown nothing yet.

ASK 19W

The tank is climbing from 4.8 towards the 7.5 limit. At what pH do you stop the base feed?

TAKES AS READ

the mixed-tank pH reading lags a pump change by one sampling interval · pH is logarithmic, so equal pH steps do not represent equal amounts of hydrogen ion

VERDICT 84W

Neutralisation in a flowing plant has delay. A pump change occurs now, but the mixed sample that proves its effect arrives later. Waiting for the displayed pH to reach the desired final value therefore guarantees overshoot. Staged dosing writes the rule in advance. The feed is slowed while buffer demand can still be measured. It is then stopped early enough that mixing and residual base carry the final reading upward without crossing the limit. The threshold is a control decision, not a retrospective interpretation.

TAKEAWAY 25W

When a process has delay, a control threshold must be set before the final limit because the system keeps moving while the action takes effect.

7.4 Water first, or never water

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM ·
CALLBACK · BELT

SCENE 36W

The contractor lifts at four and wants the yard sorted by then. Some of these drums are made safe by dilution. Others react with water, and a hose is the worst thing that could reach them.

GUIDE 46W

Two bins. Something that simply dissolves is diluted by water and made less dangerous by it. Something that reacts with water makes heat, gas or both, so water is what turns a spill into an event. Sort on what the substance does when water reaches it.

ASK 14W

Send each drum to the bin that says what water would do to it.

BEHIND THE BUTTON 91W

Why this is the first question at a yard. The default response to almost anything is water — hoses are what a fire crew has. For most of what is here that is right, and for a short list it is the one action that makes the situation worse.

What reacting with water looks like. Alkali metals and hydrides give off hydrogen. Concentrated acids give off enough heat to boil what is poured onto them. Carbides and some chlorides give off gas that is worse than what was in the drum.

TAKES AS READ

some substances dissolve in water and some react with it · atomic structure, ions and electron configuration — taken as read

VERDICT 166W

Water is the default response to a spill because it is what everybody has, and for most of a yard it is the right one: a soluble salt or an acid already in dilute solution is made less dangerous by more water, and the concentration is the hazard. The short list is where the default inverts. Alkali metals and hydrides displace hydrogen from water and give off enough heat to light it. Concentrated sulphuric acid releases so much heat on dilution that water poured onto it boils and throws the acid back — which is why the rule is acid into water and never the reverse. Carbides give off acetylene, and some chlorides give off hydrogen chloride gas that is worse in air than the drum was on the ground. Nothing about the drums looks different. The label is the only thing that tells you which list a drum is on, which is why a yard of unlabelled drums is the emergency rather than an inconvenience.

TAKEAWAY 14W

The default response is right for most things and catastrophic for a short list.

END OF THE DAY 13W

Select a controlled neutralization strategy and determine when buffer capacity will be exhausted.

Equilibrium in the Reservoir

PLAN CARD 125W

Day 14, and a fortnight of results has been clean. The dissolved amount is down, the amount in the sediment is up by about as much, and Elena Stavros, who will have to monitor it for a decade, is the only person in the room still asking about it — everybody else has moved on to the treatment plant. Today you predict where the contaminant goes the next time conditions move, and design monitoring that would catch it coming back. Nothing was destroyed by the correction. It was moved into a form that is not in the water today, and it is still in the reservoir. What tests that claim is a storm, a warm month, or the autumn week when the lake turns over.

BRIEFING 38W

After pH adjustment, dissolved contaminant decreases but sediment concentrations rise. The team must determine whether precipitation and complexation have transferred rather than destroyed the hazard. An apparently clean reservoir can become contaminated again when chemistry or flow changes.

WHAT YOU DECIDE 16W

Predict how pH and ligands shift dissolved concentration and design monitoring for the metal coming back.

8.1 From condition change to new equilibrium

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A ROOM · SEQUENCE

SCENE 35W

After the pH adjustment the dissolved concentration has fallen and the sediment concentration has risen. Reading only the water is how a reservoir gets declared clean while the contaminant settles to the bottom of it.

GUIDE 58W

Each of these four needs something the others supply. Ask what each one cannot be done without. An equilibrium expression cannot be written for chemistry nobody has named. Changing a condition means nothing until the model balances mass. And the last card is the one this stop turns on: a prediction nobody measured against is not a prediction.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

material is conserved — a shift moves it, it does not destroy it

VERDICT 83W

Predicting where a system lands takes both halves of the problem. The relevant reactions have to be identified first, because you cannot write an equilibrium expression for chemistry you have not named. The expressions and the mass balance come next: the expressions say which direction the system moves, and the mass balance says how much of what has to be somewhere. Only then does changing a condition in the model mean anything, and only then can a prediction be compared against a measurement.

TAKEAWAY 13W

A lower dissolved concentration may mean phase transfer, not removal from the system.

8.2 Interpret the shift

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A PERSON · PROTOCOL

SCENE 41W

Reservoir conditions are changing as the treatment continues. Operators have removed some solids, a new ligand has entered with an upstream discharge, and pH varies through the day. Next spring, the city will not have crews sampling continuously for remobilised metal.

GUIDE 71W

Four changes to the reservoir, and each pushes the same equilibrium a different way. Ask of each which side it takes something from or adds something to. One of them raises the total dissolved metal while lowering the free metal, which reads as a treatment failure and is not. Next spring nobody will be sampling continuously, so the answer has to be about the environment the sediment will be in then.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a system at equilibrium responds to a change by partly offsetting it

VERDICT 83W

Le Chatelier's principle gives the direction, and each condition pushes a different way. Removing a solid phase from contact takes away the source of further dissolution. A ligand that binds the dissolved metal holds it in solution, so total dissolved concentration can rise even as free metal falls. A pH shift that favours an insoluble hydroxide drives precipitation. A sediment that later meets acidic water

can release what it captured. A contaminant locked into a solid this month can therefore return next month.

TAKEAWAY 12W

Treatment plans must anticipate the next environment, not only the current sample.

8.3 Predict whether the salty pond freezes

SOLUTIONS & WATER CHEMISTRY • WATER INTAKE LABORATORY • A PERSON • BALLPARK

SCENE 44W

The holding pond is at $-4\text{ }^{\circ}\text{C}$ while the freshwater ditch beside it is frozen. Analysis gives 1.2 mol of dissolved NaCl per kilogram of water. Operators need to know whether the liquid pond is chemically expected or evidence of an unmeasured heat source.

GUIDE 71W

Five numbers, and two of them are reference points rather than terms. One is the freezing point of pure water; one is the temperature the pond is at. Another is the number of particles each formula unit contributes, which is where the salt does its work. Ask of each whether the relationship needs it or the comparison does. The answer decides whether this is ordinary chemistry or an unmeasured heat source.

ASK 9W

Should this pond still be liquid at $-4\text{ }^{\circ}\text{C}$?

BEHIND THE BUTTON 121W

What the relationship says. $\Delta T_f = iK_f m$, and $T_{\text{freeze}} \approx 0\text{ }^{\circ}\text{C} - \Delta T_f$.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

molality is moles of solute per kilogram of solvent • NaCl contributes about two dissolved particles per formula unit in this estimate

VERDICT 75W

For an ideal solution, $\Delta T_f = iK_f m$. Sodium chloride contributes roughly 2 dissolved ions, so i is about 2. With $K_f = 1.86\text{ }^{\circ}\text{C kg/mol}$ and $m = 1.2\text{ mol/kg}$, the depression is about $4.5\text{ }^{\circ}\text{C}$. That puts the expected freezing point near $-4.5\text{ }^{\circ}\text{C}$, so liquid water at $-4\text{ }^{\circ}\text{C}$ is plausible without any extra heat source. The estimate is imperfect at high salt concentration, but it gives the correct scale and direction.

TAKEAWAY 21W

Freezing-point depression depends on the number of dissolved particles, so concentration can predict whether a solution remains liquid below $0\text{ }^{\circ}\text{C}$.

8.4 Interpret plume behavior

ATMOSPHERE & GASES · AIR & PLUME STATION · A ROOM · CALLBACK · PROTOCOL

SCENE 38W

The weather station shows rising temperature, falling pressure and a strengthening afternoon wind while the plume is still moving. Ground crews also report vapour pooling in a low drainage channel on the downwind side of the freight yard.

GUIDE 69W

Two different physics run at once here, and the four situations do not all belong to one. Some change what a fixed amount of gas occupies. Others change where the parcel goes. Ask which of the two each one acts on before pairing it. Read a transport change as a gas-law change and the corridor is drawn from the wrong physics. The vapour is pooling in a drainage channel.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a gas has a state and also a place, and they change for different reasons

VERDICT 85W

Two different physics are running at once. Gas-law behaviour says how a fixed amount of substance responds to temperature and pressure. Charles and Boyle are two statements about the same gas, one proportional and one inverse. Atmospheric transport says where the parcel actually goes. Density relative to air affects whether the cloud hugs the ground or lifts. Wind and turbulence control dilution, while terrain controls where vapour can pool. Each change on the board pushes on one part of that system, not necessarily the others.

TAKEAWAY 9W

A plume is a coupled thermodynamic and transport problem.

END OF THE DAY 16W

Predict how pH and ligands shift dissolved concentration and design monitoring for the metal coming back.

The Air Is Not Clear Yet

BEFORE THE DAY — CANVASS

Who actually drank the tap water

The advice went out on Wednesday and the exposure estimate depends on whether people followed it. Ask along the affected streets until enough answers have come back to tell a city that complied from a city that did not, because those two produce very different numbers and the same reassuring press line.

PLAN CARD 101W

Day 17. The main plume cleared a week ago and residents came home the day after. Since then Varga's monitors climb every afternoon and drop overnight, and he has been ignored twice because the release is over. He has three fixed monitors, all of them on the yard fence, and one van. Today you work out what is forming out there and move the monitoring to where and when it could see it. Monitoring built for the release will keep reporting that the release is over, and it will be right. People are back in their houses with the windows open.

BRIEFING 40W

Sunlight and atmospheric oxidants transform remaining vapors into secondary products. The player must reason about radicals, reaction pathways, and changing exposure after the source is controlled. Residents may return after the primary plume clears while secondary pollutants are still forming.

WHAT YOU DECIDE 10W

Identify likely secondary products and update monitoring locations and times.

9.1 Build a secondary-pollutant pathway

ATMOSPHERE & GASES · AIR & PLUME STATION · A ROOM · SEQUENCE

SCENE 38W

The primary vapour is falling after the source was sealed, but a different compound is rising at the neighbourhood monitor each afternoon. Sunlight, oxidant measurements and wind data from the same period are posted beside the chemical traces.

GUIDE 65W

These four are a chain, and each step makes what the next one needs. Ask what has to be present before each card can happen. Radicals cannot attack anything before something creates them. Nothing can be transported before it exists. The primary vapour is falling while a different compound rises each afternoon. The order of this chain tells the monitoring team which compounds to add.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

sunlight carries enough energy to break chemical bonds

VERDICT 79W

Sunlight can start atmospheric chemistry that does not occur at night. Photons break bonds or create reactive radicals, and those radicals rapidly attack other molecules in air. The products can differ in toxicity from what was emitted. A primary vapour can therefore fall while a secondary pollutant rises later and farther downwind. Building the pathway in order links the daytime pattern to a mechanism. It also tells the monitoring team which new compounds and chemical drivers should be measured.

TAKEAWAY 10W

Exposure can peak after the original emission begins to decline.

9.2 Read the day-night pattern

ATMOSPHERE & GASES · AIR & PLUME STATION · A PERSON · DIAGNOSIS

SCENE 42W

The source has been sealed, but the afternoon monitor is climbing again. The shelter notice expires today unless the health desk renews it. Two stations have run through a full day, and their traces do not look like a plume drifting away.

GUIDE 70W

Six readings, and the inert tracer and the shared inlet material are there to rule two candidates out. Compare the whole day at both stations rather than the afternoon at one. A plume that drifted back carries its own tracer with it; an inlet artefact appears at both monitors together. A compound made in the air has a shape: it rises with the sun, lags the maximum, and falls overnight.

ASK 26W

The source is controlled and exposure went up. Which explanation fits the whole day at both monitors well enough to renew or lift the shelter advice?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a compound blown around by wind has no reason to care what time it is

VERDICT 75W

Photochemistry has a signature. Production needs sunlight, so a compound formed in air rises during the day. The peak can lag the solar maximum because the chemistry takes time. Production then falls overnight as the reactive radicals disappear. Formation in the air also shifts the peak downwind because the parcel reacts while it travels. A compound that is merely transported lacks this coordinated day-night and downwind pattern. That makes the shape diagnostic rather than incidental.

TAKEAWAY 11W

Atmospheric chemistry must be inferred from coordinated chemical and weather patterns.

9.3 Monitor the transformed plume

ATMOSPHERE & GASES · AIR & PLUME STATION · A PERSON · VALUE

SCENE 40W

Residents want to come home. The released compound is down to 3 parts per billion at both monitors. Total oxidant is up fourfold since Tuesday, and afternoon readings remain highest even though the source has been sealed for six days.

GUIDE 48W

Two channels can be added this week, and residents want to come home. Open each measurement and ask whether its result could change that decision. The released compound is already low at both monitors, so more of the same measurement cannot move anything. Buy the channels that can.

ASK 9W

Which measurements can change the return-home decision this week?

BEHIND THE BUTTON 135W

Why the original compound is no longer the question. It is down to a few parts per billion and the source has been sealed for six days. A measurement that confirms a number everybody already accepts cannot change a decision, however cleanly it is made.

What the oxidant rise is saying. Total oxidant up fourfold, with afternoon readings highest, is the signature of something being formed in the air rather than released from the site. Sunlight drives it, which is why it peaks in the afternoon and why sealing the source did not stop it.

What a return-home decision actually needs. Not the concentration of the thing that was spilled, but of whatever residents will breathe next week. Those are different molecules, and the monitors currently watching for the first are silent about the second.

TAKES AS READ

a monitoring network only detects the compounds it is configured to measure · chemical drivers can explain exposure without being the exposure themselves

VERDICT 84W

The source compound is already low at both existing monitors, so measuring it more precisely does little to test the new hazard. Sunlight and oxidants explain why secondary chemistry should occur, but they are not themselves the exposure. The decisive channel measures the predicted secondary products where people breathe them. A second channel can then measure the chemical drivers to explain and forecast the pattern. This ordering separates evidence that changes the return-home decision from evidence that merely makes the mechanism easier to describe.

TAKEAWAY 21W

When chemistry transforms a released compound, monitoring must follow the exposure product rather than only the original source or its drivers.

9.4 From identity to reaction estimate

REACTIONS & ENERGY · REACTION & THERMAL HALL · A ROOM · CALLBACK · SEQUENCE

SCENE 41W

At the tunnel entrance, two drainage streams may be meeting behind a closed access door. A repair crew is waiting at the barricade, and the storm bypass remains shut. The laboratory has analyses from both drains but no reaction estimate yet.

GUIDE 66W

These four are not four independent tasks. Each needs the output of another. Ask of each card what it requires before it can be done at all. Coefficients cannot be used until the equation is balanced. Masses cannot be compared until they are moles. A prediction is worth making only once there is a measurement to compare it against. A repair crew is at the barricade.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a balanced equation says how many of one thing react with how many of another

VERDICT 74W

Every quantitative reaction prediction begins with a balanced equation. Its coefficients supply the mole ratios between reactants and products. Measured masses are then converted to moles, because the coefficients compare particles rather than grams. Those mole ratios identify which reactant runs out first. Only after the limiting reactant is known can the maximum products be calculated. The prediction can then be checked against the gas, heat or remaining reagent actually observed in the tunnel.

TAKEAWAY 10W

Stoichiometry is a conditional prediction based on a specified reaction.

END OF THE DAY 10W

Identify likely secondary products and update monitoring locations and times.

The Heat Is Rising

PLAN CARD 103W

Day 21, eleven in the morning. The fire has been out for three weeks and one storage bay is still getting warmer. Tomas Brandt, who owns reactions and energy, has been reading it hourly since midnight and has 11 readings. Fire Command wants the site released tonight and the crews have been on shift for 14 hours. Today you work out how much energy it takes to warm that much material, decide whether you are watching leftover heat or a reaction still running, and set up the monitoring either way. Fire Command wants your answer after the next reading, not after the trend.

BRIEFING 42W

A storage zone continues warming after the visible fire is out. The team must distinguish stored heat from ongoing exothermic reaction and estimate whether cooling can keep up. If self-heating is mistaken for leftover warmth, responders may leave before runaway conditions develop.

WHAT YOU DECIDE 13W

Create an energy balance and choose a monitoring plan that detects self-heating early.

10.1 How much energy raises the temperature?

REACTIONS & ENERGY · REACTION & THERMAL HALL · A ROOM · BALLPARK

SCENE 41W

The visible fire has been out for two hours, but the containment bath beside the damaged storage bay is still warming. Fire Command wants the size of the stored heat load before deciding whether the cooling equipment on site is adequate.

GUIDE 68W

Five numbers, and two of them belong elsewhere: one to a change of phase, one to the surrounding air. Ask of each whether it describes the mass, the material, or the change. Latent heat answers a different question and is much the largest number here. A small warming across a large thermal mass is a great deal of energy. Fire Command is sizing cooling equipment from the answer.

ASK 4W

Estimate the absorbed heat.

BEHIND THE BUTTON 130W

What the relationship says. $q = mc\Delta T$, with q the heat, m the mass, c the specific heat capacity and ΔT the temperature change.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

different materials need different amounts of energy to warm by one degree

VERDICT 77W

The energy needed to change a material's temperature is its mass times its specific heat capacity times the temperature change. A small rise across a large thermal mass can therefore represent a very large amount of energy. Water's high heat capacity makes it a useful containment bath, but it also hides how much energy has entered. Temperature alone says little about the quantity of stored heat. It becomes useful only when connected to mass and heat capacity.

TAKEAWAY 10W

Temperature becomes meaningful when connected to mass and heat capacity.

10.2 Stored heat or continuing reaction?

REACTIONS & ENERGY · REACTION & THERMAL HALL · A PERSON · DIAGNOSIS

SCENE 39W

Two storage bays were exposed to the same fire. 90 minutes after cooling stopped, the east bay is cooling while the west bay is warming. West-bay off-gas is increasing, and the hottest point remains fixed at the drum stack.

GUIDE 71W

Six readings are on the panel and the east bay is the useful one, because it is the same fire with nothing left running. Compare the two bays reading by reading before choosing. A miscalibration moves both bays together; stored heat can only drain toward cooler surroundings. Rising temperature with rising off-gas at one fixed point is a different claim from either, and Fire Command releases the site on this answer.

ASK 24W

Fire Command needs to know whether this is heat left over from the fire or a reaction still running. Which explanation fits every zone?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

heat flows from hot to cold and nowhere else on its own

VERDICT 82W

A hot mass can be carrying stored energy from an earlier event, or it can be generating new energy now. After the source is removed, stored heat can only drain toward cooler surroundings. The east bay shows that pattern and provides a control. The west bay instead rises in temperature while off-gas increases at the same fixed location. That combination requires ongoing heat generation. A calibration error

would move both bays together, so the diverging trends point to chemistry rather than instrumentation.

TAKEAWAY 11W

Thermal diagnosis requires an energy balance, not a single temperature threshold.

10.3 Control the self-heating risk

REACTIONS & ENERGY · REACTION & THERMAL HALL · A PERSON · CHOICE

SCENE 36W

Fire Command wants the site released tonight. Bay 3 has risen from 31 to 44 degrees in 8 hours, the rate of rise is itself increasing, and the mass is 4 tonnes with a two-metre depth.

GUIDE 67W

All four options are things somebody could require tonight, and three of them are genuinely useful. Ask of each what it would tell you that you do not already know, and how soon. Self-heating is a race between heat made and heat carried away, so only a trend says which side is winning. One reading at the door cannot show a trend, however good the reading is.

ASK 12W

Fire Command wants the site released tonight. What do you require first?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a trend says something a single reading cannot

VERDICT 83W

Self-heating is a race between the energy a reaction makes and the energy the surroundings can carry away. Only the trend says which side is winning, so a site is safe to release when somebody can watch the trend — not when one reading happens to look fine. Controlling it takes three things that do not stand in for each other: detection early enough to act on, the mechanism so the action is the right one, and enough cooling to shift the balance.

TAKEAWAY 8W

High-consequence thermal systems need detection, understanding, and mitigation.

10.4 pH seven, while the plant keeps feeding

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A ROOM · HOLD

SCENE 41W

The neutralisation tank has to sit at pH 7 before anything leaves for the river. Acid liquor arrives from the works in batches, the lime dosing pump is the only control, and the analyser on the outfall reads every few seconds.

GUIDE 57W

Hold the tank at pH 7 inside the band on the analyser. The band narrows as the discharge window approaches, because a consent is written on what leaves rather than on what was in the tank an hour ago. The lime pump is your control, and each batch that arrives keeps pushing until the dosing answers it.

ASK 12W

Hold the tank at pH 7 while the works keeps feeding it.

BEHIND THE BUTTON 143W

Why pH is not a quantity you can chase. It is a logarithm, so the same dose of lime moves the tank far more near neutral than it does at pH 3 — the last half unit is where a plant overshoots and finds itself dosing acid back in.

Why the buffer decides how twitchy this is. Water with carbonate in it resists change: the same acid batch moves a buffered tank much less. When the buffer is used up the tank stops resisting and moves fast, which is what an operator means by the tank "going".

Why a batch is not a step in the reading. Liquor arriving is a rate, and it keeps arriving. A tank left alone does not settle at a new pH — it carries on moving until the dosing is set to match what is coming in.

TAKES AS READ

pH measures how acidic or basic a solution is

VERDICT 162W

Each batch here is a step in the rate rather than in the reading. Acid liquor arriving does not drop the tank by a tenth of a unit and stop; it keeps arriving, and the tank keeps falling until the dosing is set to match it. That is why the lime pump is moved to a new rate and held. Two things make pH harder than a temperature. It is logarithmic, so the last half unit near neutral takes a fraction of the lime the first three units took, and an operator who has learned the feel of the tank at pH 4 will overshoot at 6.8. And the buffering changes under you: while there is carbonate in the water the tank resists, and once it is consumed the same batch moves it several times as far. The band narrowing is the consent — what is measured is what leaves the works, not what the tank was doing when the shift started.

TAKEAWAY 12W

A feed you do not answer is not a feed that stops.

10.5 Why the drum is worse in the afternoon

ATMOSPHERE & GASES · AIR & PLUME STATION · A ROOM · CALLBACK · CHOICE

SCENE 35W

The same open drum reads three times the vapour concentration at two in the afternoon as it did at six in the morning. Nothing has been added to it and nothing has been taken out.

GUIDE 72W

All four options would explain a rising reading, and they differ in what has changed. One changes the liquid, one changes the instrument, one changes the amount, and one changes only the partition between liquid and air. Nothing was added and nothing was removed, which is the fact that does the work. Getting this wrong sets an exposure limit from a dawn reading and clears a site that is worse by mid-afternoon.

ASK 9W

Same drum, same liquid, three times the vapour. Why?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a liquid and its vapour sit in equilibrium at a temperature

VERDICT 88W

Vapour pressure is the pressure of gas in equilibrium with its own liquid, and it climbs steeply with temperature — often doubling for every 10 or 15 degrees. Warming the drum from morning to afternoon moves more molecules into the gas phase above the same liquid, so the air above it is more concentrated with no change in what is in the drum. This is why exposure limits are checked at the hottest part of the day, and why a hazard assessment made at dawn understates the afternoon.

TAKEAWAY 18W

How much of a liquid is in the air above it depends steeply on how warm it is.

END OF THE DAY 13W

Create an energy balance and choose a monitoring plan that detects self-heating early.

The Corrosion Failure

PLAN CARD 92W

Day 26. A section of the intake pipeline has failed and Novák is standing in the hole in Third Street. It is also the main your own team clamped an earth cable to a fortnight ago. That pipe carries the water 140000 people drink, and the city is already on its reserves. Today you find where the metal is being lost and why, work out how fast, and choose the protection. Speed decides your response: the same loss over ten years is maintenance, and over a fortnight it is a burst main.

BRIEFING 36W

Acidic water and different metals threaten the intake pipeline. The player must identify redox couples and stop a small leak from becoming a system-wide failure. A hidden localized failure could interrupt drinking water during the emergency.

WHAT YOU DECIDE 14W

Identify the corrosion cell and choose controls that address both chemistry and electrical coupling.

11.1 Find the anode and cathode

TREATMENT & INFRASTRUCTURE • PILOT TREATMENT PLANT • A ROOM •
PROTOCOL

SCENE 40W

At the intake gallery, acidic water is flowing through a section where two different metals are electrically connected. A coating inspection has also found one small breach. The maintenance team wants the corrosion circuit identified before anyone chooses a repair.

GUIDE 64W

A corrosion cell needs four things at once. Somewhere metal is lost. Somewhere electrons are consumed. A path for the electrons, and a path for the ions. Each situation on the left supplies or concentrates one of those parts. Ask which part before pairing anything. Break any one and the circuit stops. So naming the parts is what decides which repair is a repair.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

oxidation is losing electrons and reduction is gaining them • atomic structure, ions and electron configuration — taken as read

VERDICT 80W

Corrosion is an electrochemical circuit with four required parts. The anode is where metal is oxidised and lost. The cathode consumes the released electrons in a reduction reaction. Electrons also need a conducting path through the metal, while ions need a path through the water. Breaking any one part stops the circuit. Two

different metals connected in an electrolyte can supply all four. A small coating breach can then concentrate anodic current into one patch and drive rapid local attack.

TAKEAWAY 9W

Stopping corrosion requires breaking or controlling the full circuit.

11.2 How fast can material disappear?

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · BALLPARK

SCENE 39W

A pipe coupon from the intake gallery has lost 2.0 kilograms of steel per year across 4.0 square metres of exposed surface. The steel density is 7,900 kilograms per cubic metre, and the remaining wall is 6 millimetres thick.

GUIDE 67W

Six numbers, and one of them is the density of water rather than of steel. Another is the wall that is left, which belongs to the comparison and not to the rate. Ask of each whether the relationship needs it or the decision does. Notice what a uniform rate cannot see. The same metal loss concentrated into one coating defect perforates far sooner than the average suggests.

ASK 8W

What uniform wall-loss rate does 2.0 kg/year imply?

BEHIND THE BUTTON 129W

What the relationship says. $\text{depth loss rate} = \text{mass loss} / (\text{density} \times \text{exposed area})$. Convert metres per year to millimetres per year.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a rate becomes a decision only when it is compared against a thickness · atomic structure, ions and electron configuration — taken as read

VERDICT 80W

Work out the average wall loss because it sets the scale of the problem. 2 kilograms per year over 4 square metres of steel is about 0.063 mm of depth per year. At that uniform rate, losing a 6 mm wall would take roughly 95 years. That result says uniform corrosion is not the immediate threat. Localised attack is worse because the same total metal loss can be concentrated into a small coating defect and perforate the wall far sooner.

TAKEAWAY 16W

A corrosion rate becomes a decision only once it is a depth compared against a wall.

11.3 Protect the pipeline

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · CHOICE

SCENE 38W

The thinnest section of intake pipe has only 3 millimetres of wall left. Corrosion has accelerated eightfold, and the maintenance chief has cleared the gallery until the first control is chosen. Water is still moving through the line.

GUIDE 62W

Three of the four options act on a real part of the corrosion circuit. One only moves paper. Ask of each which of the four parts it removes, and whether one small defect can defeat it. The water is still moving and three millimetres of wall are left. So the first control has to break the circuit, not merely slow it down.

ASK 18W

The gallery stays cleared until the first control is chosen. Which intervention breaks the corrosion problem most directly?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

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TAKES AS READ

a circuit stops when any one part of it is broken · atomic structure, ions and electron configuration — taken as read

VERDICT 77W

A corrosion cell needs an anode, a cathode, an electrolyte and an electrical path. The acidic, chloride-rich water makes ion transport easy. The connected dissimilar metals provide an electron path and drive galvanic attack. Electrical isolation breaks that path directly. A coating can help, but one small defect can concentrate attack. Chemistry control also matters, yet it does not remove the metal-to-metal connection. The first intervention should break the circuit while inspection and water chemistry controls follow.

TAKEAWAY 11W

Corrosion protection is a designed system of materials, environment, and inspection.

11.4 Which reactant limits?

REACTIONS & ENERGY · REACTION & THERMAL HALL · A ROOM · CALLBACK · BALLPARK

SCENE 32W

A reaction takes two moles of A for every mole of B, and both are arriving in the tunnel in amounts nobody chose. Closed-space entry decisions are being made from this number.

GUIDE 65W

Four numbers, and one of them is the two reactant amounts added together. Another is the reaction's own ratio rather than a quantity of anything. A balanced equation compares particles, so amounts have to be in moles before the ratio can be applied. Adding two reactants gives a number that no stoichiometry uses, and closed-space entry decisions are being made from whichever number comes out.

ASK 14W

Which reactant limits, how much B can react, and what percent yield was recovered?

BEHIND THE BUTTON 133W

What the relationship says. First use $n = m/M$. Then compare mole amounts with the 2 A : 1 B coefficients. Percent yield = actual/theoretical $\times$ 100.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a reaction stops when one ingredient runs out

VERDICT 72W

A balanced equation compares particles, so masses have to become moles before the reactants can be compared. 500 grams of A at 50 g/mol is 10 mol. 640 grams of B at 80 g/mol is 8 mol. The 2:1 ratio means 10 mol A can consume only 5 mol B, so A limits. If the recovered product corresponds to 4.2 mol of reaction, the percent yield is $4.2/5.0 \times 100 = 84\%$.

TAKEAWAY 17W

Stoichiometric limits are set in moles, and percent yield compares what was recovered with that theoretical limit.

Identify the corrosion cell and choose controls that address both chemistry and electrical coupling.

Will the Reaction Run Away?

BEFORE THE DAY — EVADE

The site owner who wants a verbal all-clear

The yard's owner has been at the gate since seven wanting somebody to say the drums are inert, and Nakamura, the quality assurance lead, has been plain that nothing is said before the assay. Keep out of his way and finish your round. An all-clear given at a gate is an all-clear with no laboratory behind it.

PLAN CARD 103W

Day 31. The treatment chemical sat quietly in a drum all week; in Brandt's warm pilot vessel it is lively. His own trace shows an hour of very little and then a step. Osei plans to run it at full scale on Monday — the pilot vessel holds 40 litres and the plant tank holds 9000. Today you decide what temperature range this process may run in, and what shuts it down without asking anybody. Treatment also gathers a spill spread thin across a yard into a single tank, and if that tank runs away, the whole of it leaves at once, hot.

BRIEFING 40W

A treatment chemical reacts slowly at room temperature but rapidly as the vessel warms. The team must connect rate laws and activation energy to thermal feedback. A treatment intended to remove contamination could become the largest hazard on the site.

WHAT YOU DECIDE 15W

Identify the controlling rate factors and define a safe operating envelope with automatic shutdown triggers.

12.1 What changes the rate?

REACTIONS & ENERGY · REACTION & THERMAL HALL · A ROOM · PROTOCOL

SCENE 38W

The same treatment mixture is quiet in a cool drum and vigorous in the warmed pilot vessel. On the bench are concentration logs, temperature records and a catalyst trial, all taken from the same batch before scale-up begins.

GUIDE 71W

The four changes on the left act at three different places. One is the concentration term. One is the fraction of collisions that clear the barrier. One is the height of the barrier itself. Ask which of the three each change touches before pairing it. One of them changes the thermodynamics and leaves the barrier alone. That is the difference between quiet in a drum and vigorous in a warm vessel.

ASK 10W

Which change affects collision frequency, barrier crossing, or activation energy?

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

kinetics means how fast a reaction proceeds and what controls that rate · a reaction needs molecules to meet with enough energy to react

VERDICT 76W

Rate laws separate concentration effects from the rate constant. If $\text{rate} = k[A]^2$, doubling $[A]$ multiplies the rate by four because the concentration term is squared. Temperature works differently: it changes the fraction of molecular encounters able to cross the activation barrier, which changes k . A catalyst also changes the pathway and lowers that barrier. Changing product energy changes thermodynamic driving force, but it does not by itself specify the kinetic barrier or the reaction rate.

TAKEAWAY 15W

A reaction can be favorable yet slow, or fast only after a barrier is lowered.

12.2 Build a runaway feedback loop

REACTIONS & ENERGY · REACTION & THERMAL HALL · A PERSON · SEQUENCE

SCENE 39W

The pilot vessel showed a sharp temperature rise during one treatment run. Four events from the log are posted on the control-room board. They have been printed out of order, and the engineering team needs the causal sequence reconstructed.

GUIDE 70W

These four are one loop, so every card has something before it and something after it. Ask of each what has to be true already for it to happen. Heat has to exist before it can raise a temperature, and a temperature has to rise before a rate constant does. Put the chain in order and a run that looked alarming becomes something a cooling system can be sized against.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a reaction goes faster when it is hotter

VERDICT 75W

It is a feedback loop with an obvious first step. An exothermic reaction releases heat. That heat raises the temperature when removal cannot keep pace. The cooling system removes heat at a rate set by its geometry and the available temperature difference. A higher temperature raises the rate constant. A faster reaction releases the same heat sooner, and the loop closes. Put the chain in order and it stops being alarming and starts being designable.

TAKEAWAY 11W

Runaway risk emerges from coupling reaction rate to imperfect heat removal.

12.3 Define the safe operating envelope

REACTIONS & ENERGY · REACTION & THERMAL HALL · A PERSON · STRESS

SCENE 40W

The first full-scale treatment batch is scheduled before dawn. Bench work stopped at 55 °C, the proposed vessel runs at 60 °C, and the cooling jacket tops out at 40 kilowatts. Heat release above 65 °C has never been measured.

GUIDE 64W

Move the uncertain doubling interval to its pessimistic end before you commit to anything. A plan that survives only at the nominal value is not a plan. The jacket removes 40 kilowatts and no more, bench work stopped at 55 °C, and nothing above 65 °C has ever been measured. Choose the operating plan that still holds at the bad end of the range.

ASK 14W

Which operating plan remains feasible when the reaction accelerates faster than the nominal assumption?

BEHIND THE BUTTON 137W

What the doubling interval means. Reaction rate rises with temperature, and the interval is how many degrees it takes to double. A small interval means the heat release accelerates sharply, so a vessel that is comfortable at 60 °C can be uncontrollable at 68 °C.

Why the jacket is the hard limit. Cooling capacity is fixed at 40 kilowatts. Once the reaction produces heat faster than that, the temperature rises, which makes it produce heat faster still. That is a runaway, and no amount of operator attention reverses it once it starts.

Why the untested region is the whole problem. Above 65 °C there is no measurement at all, so the model there is extrapolation. Committing a full-scale batch to a region nobody has measured is the decision this stop is asking you not to make quietly.

TAKES AS READ

heat generation tracks reaction rate over these compared operating points · a plan is infeasible when heat generation exceeds available cooling

VERDICT 82W

Scale-up cannot be based only on the nominal rate. Heat generation rises steeply with temperature, while heat removal is capped by the jacket and its temperature difference. A plan that barely fits the nominal kinetics can cross the cooling limit if the true temperature sensitivity is stronger. Stressing the uncertain doubling interval exposes that fragility before operation. The conservative plan gives up throughput. It is the only candidate that still has positive cooling margin at the pessimistic end of the stated range.

TAKEAWAY 23W

A safe operating choice is one that still satisfies the thermal constraint when a key kinetic assumption is wrong in the dangerous direction.

12.4 What the position in the table tells you

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM ·
CALLBACK · CHOICE

SCENE 34W

Two unlabelled drums, one holding a sodium salt and one a caesium salt of the same anion. The bench has to say which will be the more violent with water before either is opened.

GUIDE 66W

Two drums share an anion, so the anion cannot separate them. Each option offers a different basis for the ranking: position in a group, atomic mass, the shared part, or refusal to predict. Ask what changes down group 1, and why. The table exists to rank two substances a bench has never handled. That ranking is what a crew works to before either drum is opened.

ASK 12W

Which drum is the more reactive with water, and on what grounds?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

the periodic table groups elements that behave alike · atomic structure, ions and electron configuration — taken as read

VERDICT 79W

Group 1 reactivity rises going down: the outer electron sits further from the nucleus and is more shielded, so it is lost more easily. Caesium is below sodium, so the caesium salt is the more energetic with water. The anion is the same in both, which is exactly why the cation decides it. This is what the table is for — it lets a bench rank two substances it has never handled, from position alone, before anything is opened.

TAKEAWAY 11W

An element's position in the periodic table predicts how it behaves.

END OF THE DAY 15W

Identify the controlling rate factors and define a safe operating envelope with automatic shutdown triggers.

Remove It Without Making It Worse

PLAN CARD 100W

Day 38. Osei's pilot treatment works — 98% of the compound you identified is gone by the outlet — and he wants to scale it this week. Nakamura points out that the panel it was tested against was written for that compound and nothing else. Today you work out what the treatment actually did: what it removed, what it made, and where the rest went. A test panel written for the original reports success either way, because it is not looking for what you made. The hazard that lasts longest here may be one there is still no test for.

BRIEFING 40W

Three treatment methods can lower the target contaminant, but one creates a toxic byproduct and another shifts the contaminant into sludge. The player must compare complete chemical consequences. A rapid treatment can create a less visible but more persistent hazard.

WHAT YOU DECIDE 14W

Choose a treatment train using contaminant removal, byproduct formation, waste fate, and operational reliability.

13.1 What did the treatment actually do?

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A ROOM · BALANCE

SCENE 39W

The pilot treatment processed a batch containing 100 kilograms of contaminant. Treated water, sludge and off-gas have each been measured. City officials are waiting to hear whether the process destroyed the material or merely moved it into another reservoir.

GUIDE 59W

Read every stream first, because reading costs nothing. Then count only the masses that belong to the same contaminant ledger. The batch went in with 100 kilograms, and treated water, sludge and off-gas have each been measured. Whatever the counted streams do not account for is the missing mass, and that gap is what the destruction claim rests on.

ASK 17W

How much material is missing from the measured streams, and does the ledger support a destruction claim?

BEHIND THE BUTTON 146W

What counting a stream claims. Reading is neutral. Adding a stream to the ledger says that the mass in it is the same substance the batch went in with. A breakdown product measured by a method that reports it as the parent compound is the way that claim goes wrong.

Why a gap is not a destruction. Mass that has left the measured streams has gone somewhere: an unmeasured vent, adsorbed on the vessel, or transformed into something none of the three methods sees. Destruction is one explanation for a gap, and the least testable of them.

What the city is actually being told. 'Destroyed' and 'moved into another reservoir' have very different consequences for whoever ends up with the sludge or the air. The ledger is what distinguishes them, which is why the claim has to be made from the numbers rather than about them.

TAKES AS READ

mass is conserved across the treatment process · a concentration decrease in water does not identify where the removed material went

VERDICT 77W

The treated water contains only 6 kg, but that does not mean 94 kg were destroyed. 71 kilograms are measured in sludge and 3 kg appear in off-gas. Those visible streams total 80 kg, leaving a 20 kg term required to close the original 100 kg ledger. The bench already sees an unquantified transformation-product signal, so that missing mass has a plausible destination. The mass balance supports transfer and transformation, not a claim that the contaminant vanished.

TAKEAWAY 19W

Removal from one stream is not destruction; a treatment claim must conserve mass across all measured products and reservoirs.

13.2 Select a treatment train

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · SEQUENCE

SCENE 30W

Four unit processes are on the table and the plant wants them in an order. Riverton has one intake and cannot run this experiment twice while the water is off.

GUIDE 66W

These four are not independent machines. Each changes the chemical form that reaches the next. Ask of each card what has to exist before it can do anything at all. Nothing can be settled or filtered until there is a solid. There is no solid until the metal is in the right state at the right pH. Riverton has one intake and cannot run this twice.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a filter can only remove something that is already a solid

VERDICT 77W

A treatment train is not four independent machines. Each stage changes the chemical form reaching the next stage. Oxidation must first create the metal state that can form an insoluble hydroxide. The pH adjustment then creates that solid. Only after a solid exists can settling and filtration remove it. Activated carbon belongs last as a polishing step for material that remains dissolved. Sending suspended solids to carbon too early can blind the bed and disable everything downstream.

TAKEAWAY 16W

A treatment train is ordered by what each stage leaves the next one to work with.

13.3 Choose the safest pilot program

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · CHOICE

SCENE 39W

The pilot has room for one question, four weeks, and a tenth of the full-scale flow. Three of the four proposals depend on the same unmeasured quantity, and the full plant is committed in June whatever the pilot says.

GUIDE 69W

Every method on the table removes the target compound, which is why all of them are on the table. So a pilot measuring removal cannot see the difference it was run to find. Ask of each option what it would tell you that the others would not, and whether it would change the June decision. Judged on removal alone, the winner is whichever method makes the most byproduct fastest.

ASK 12W

All three methods claim removal. What does the pilot have to measure?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a method that removes a compound has to put it somewhere

VERDICT 88W

Every method on the table removes the target — that is why they are on the table. What separates them is what they leave behind: transformation products, residuals, and the lifecycle of the waste. A pilot that measures only the target compound cannot see the difference it was run to find, and judged on removal alone it will pick whichever method makes the most byproduct fastest. The right question is the one that would change the decision, and here that means analysing for the products each mechanism predicts.

TAKEAWAY 11W

The best treatment minimizes total hazard, not merely one measured concentration.

13.4 Heat release is not the spontaneity test

REACTIONS & ENERGY · REACTION & THERMAL HALL · A ROOM · CALLBACK · BALLPARK

SCENE 43W

A clean-up reagent is staged beside the tunnel, but the tank test has shown no product after 6 days at 18 °C. Its measured ΔH and ΔS are on the clipboard.

Operations wants to know whether the chemistry is impossible or simply slow.

GUIDE 69W

Four numbers, and two of them are the same temperature written in different scales. The relationship needs an absolute temperature, so degrees Celsius put straight into it is not a small error but a different quantity. Note what the answer does and does not settle: a favourable sign says the reaction can go, not that it will go this week. The tank has had six days and shows nothing.

ASK 18W

At 18 °C, what is ΔG , and what does the lack of product say about the reaction rate?

BEHIND THE BUTTON 119W

What the relationship says. $\Delta G = \Delta H - T\Delta S$, with T in kelvin.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

temperature in $\Delta G = \Delta H - T\Delta S$ must be in kelvin · a negative ΔG means the forward reaction is thermodynamically favored

VERDICT 70W

Free energy combines enthalpy, entropy and temperature. At 291 K, $\Delta G = -180 - 291(-0.40) \approx -64$ kJ/mol, so the reaction is thermodynamically favorable. That does not make it fast. The unchanged tank points to a substantial activation barrier at this temperature. This distinction matters in treatment design. Temperature or a catalyst can change reaction rate without changing the sign of ΔG . A favorable ΔG alone never guarantees useful speed.

TAKEAWAY 14W

Free energy sets thermodynamic favorability; activation barriers set how quickly a favorable reaction proceeds.

Choose a treatment train using contaminant removal, byproduct formation, waste fate, and operational reliability.

Can the Water Be Released?

PLAN CARD 94W

Day 45, and Nakamura gets her day. The verification results are in and one sits just under the limit — and it is the re-run of a sample whose first quality check failed. She has the instrument's own repeatability from 40 duplicate runs and would like somebody to look at it. Reyes has schools and two hospitals on tankers and wants the release signed this morning. Today you decide: release, release with conditions, or hold. Nakamura's name is on that form under Reyes's. Both mistakes are real, and they are not the same size.

BRIEFING 40W

Treatment results cluster near the decision threshold. The player must account for uncertainty, detection limits, and sampling representativeness before reopening the water system. A premature release exposes the city; an unnecessary hold can deprive hospitals and homes of essential water.

WHAT YOU DECIDE 14W

Make a transparent release decision with conditional monitoring and explicit treatment of borderline results.

14.1 Reading a mass spectrum

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A ROOM · CHOICE

SCENE 42W

Two candidate compounds fit the chromatography peak equally well. Their proposed formulas differ by one oxygen atom, and the mass spectrometer has already measured the ion from that same fraction. The review board wants the tie broken before release paperwork is signed.

GUIDE 69W

The two candidates differ by one oxygen, so the question is which measurement responds to that difference. Sort the options by what each one actually measures: how fast something travels, how much of it there is, or what it weighs. Retention time and peak area are both real numbers about the wrong property. Release paperwork gets signed on this identification, and one atom is the whole of the difference.

ASK 11W

Two candidates differing by one oxygen. What settles it, and how?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a molecule's formula fixes its mass

VERDICT 80W

A mass spectrometer sorts ions by mass divided by charge. For a singly charged molecular ion, the molecular peak therefore reports molecular mass directly. 1 extra oxygen changes that mass by about 16 units, far larger than the ambiguity in the chromatographic retention time. Peak area instead measures response and abundance, not identity. The mass result supplies a different physical constraint

from the separation. That is why it can break a tie between two candidates that chromatography alone cannot distinguish.

TAKEAWAY 12W

A mass spectrum measures mass-to-charge, which distinguishes formulas that other methods cannot.

14.2 Close the verification gaps

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A PERSON · PROTOCOL

SCENE 35W

A number can be perfectly valid and still fail to answer the question that was asked. The distribution network is about to be brought back up to pressure on the strength of this evidence package.

GUIDE 65W

The left column is four separate weak points in a verification package. Each has its own remedy. Pair them by asking what specifically was not tested. Sampling only the plant leaves the endpoints untested. Sampling after long flushing measures water nobody receives. A detection limit above the release limit cannot test the standard at all. The network goes back up to pressure on this evidence.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a sample tells you about the place and the moment it was taken

VERDICT 80W

Verification has several separate weak points and each has its own remedy. Sampling only the treatment plant leaves the endpoints and vulnerable zones untested, and those are where people actually drink. Sampling after long flushing measures water nobody will receive, so the flushing has to be documented and realistic use conditions sampled. A detection limit above the release limit cannot test the standard at all. And one laboratory doing both the optimisation and the verification has marked its own work.

TAKEAWAY 15W

A compliant number is meaningful only if the sampling and method address the actual decision.

14.3 Approve, condition, or hold

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A PERSON · CLOUD

SCENE 40W

Distribution pumps are idle while the board meets. The verification result is 9 units against a limit of 10, plus or minus 2. Operators are waiting at the valves, and the city has already been given two different reopening times.

GUIDE 73W

The band on screen is every result the measurement permits, not the one most likely — nine against a limit of ten, give or take two, so part of it is over the limit. Two pieces of follow-up work are available, and each does something different to the band: watch what happens to its middle and to its width separately. Apply what you would authorise, then say whether the release criterion is met.

ASK 26W

Can the whole plausible band be brought below the limit, or only its central value? Place the bars to report the mean and its one-sigma uncertainty.

BEHIND THE BUTTON 172W

What the band is. A single number with a plus-or-minus on it is a claim about a range: any value inside is consistent with what was measured. Reporting only the middle throws that away. The board is not asking whether the best estimate is under the limit — it is asking whether a result over the limit has been ruled out.

Moving the middle is not narrowing the band. A rerun that finds a bias corrects where the band sits and leaves its width alone: the same uncertainty, somewhere else. Only new information — more samples, a better method, an independent measurement — makes the band narrower. Both are useful; only one of them can clear a limit that the width is crossing.

What is riding on the difference. Distribution pumps are idle and the city has already been given two reopening times. A decision taken on the middle alone reopens the network with a real chance of being over the limit, and the second retraction costs more than the first did.

TAKES AS READ

the displayed width is the uncertainty band used for this decision · a central value below a limit does not pass when the supported range crosses the limit

VERDICT 86W

The initial result is 9 ± 2 , so its supported range reaches to 11 and crosses the limit of 10. An independent rerun can move the centre without proving the original uncertainty was too wide. Representative sampling can narrow the range without changing the central estimate. In this authored case, neither action alone clears the boundary. Together they produce 8.6 ± 1.0 , whose upper edge is 9.6. Only then does the evidence support an unconditional numerical pass rather than a reversible policy decision under unresolved uncertainty.

TAKEAWAY 21W

Changing a best estimate and reducing uncertainty are different operations; a boundary decision depends on where the full supported range lies.

14.4 Read analytical disagreement

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM · CALLBACK · TRACE

SCENE 37W

Two methods have been run on the same extract and they do not tell the same story. A false identification here sends the plant a chemistry that could create a worse product than the one it removes.

GUIDE 50W

Open each channel to see what its result depended on. Keep the ones that stand on an independent measurement, and untick the ones whose chain runs through a shared step. Then name the step they have in common. Two methods disagreeing is information, not an error to be averaged away.

ASK 11W

Which channels remain trustworthy, and what shared source is at fault?

BEHIND THE BUTTON 143W

Why disagreement is the useful signal. Two methods agreeing can mean both are right or both inherit the same mistake. Two disagreeing means at least one assumption is wrong, and finding which is a question about what each method depended on rather than about which number looks better.

What a shared step looks like here. A calibration standard, a dilution, a column, an internal reference: anything both methods passed through. Everything downstream of it moves together if it is wrong, which is why the average of two such results is no safer than either.

What is riding on getting it right. A false identification sends the plant a treatment chemistry chosen for the wrong compound, and the product of that reaction can be worse than what it removes. Withdrawing both results is safer than averaging them and worse than naming the shared step.

TAKES AS READ

a blank contains every preparation step except the sample · independent methods do not share the same critical failure path

VERDICT 80W

The sample and duplicate chromatograms agree because they share the same preparation solvent, retention reference and detector path. Their agreement therefore does not provide two independent identifications. The blank carries almost the same 3.1-minute feature even though it contains no river sample, which points to preparation contamination. The structural spectrum uses a different measurement path and does not match the target. Together, the blank and the independent spectrum overturn the retention-time identification while leaving the rest of the run interpretable.

TAKEAWAY 16W

Agreement counts as independent evidence only when the measurements do not inherit the same failure path.

END OF THE DAY 14W

Make a transparent release decision with conditional monitoring and explicit treatment of borderline results.

Reopen the City

BEFORE THE DAY — TAG

Catch Brandt before the reactor bay is sealed

Brandt, the reactions and energy lead, has the only record of what temperature the drum bay reached overnight, which is what decides whether the reaction ran or merely warmed. He is walking to the bay and once it is sealed for the afternoon nobody is going in. The whole kinetics argument rests on a number in his notebook.

PLAN CARD 113W

Day 52. The mayor wants one word tonight and Mbeki has to put it on a screen behind him. Behind that word sits everything your teams have concluded in seven weeks, and those conclusions are not equally strong. Stavros wants the monitoring programme funded before the emergency budget closes on Friday; nobody else in the room has mentioned it. Today you take the claims one at a time — what was measured, what was worked out from it, what is still open — and decide what the city keeps paying for after everyone goes home. The sentence people quote back to you for years is usually the one you were least sure of.

BRIEFING 44W

The final board must explain what was released, how chemicals moved and changed, which treatments worked, and what contamination remains in solids, infrastructure, and ecosystems. Without a complete fate and uncertainty record, the city cannot know whether the emergency is over or merely hidden.

WHAT YOU DECIDE 11W

Deliver a claim-by-claim chemical evidence package and a long-term monitoring plan.

15.1 Disposition the final chemical claims

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A ROOM ·
PROTOCOL

SCENE 31W

The city wants one sentence and the chemistry does not support one. Collapsing everything into a single green light is how a site gets reoccupied above a problem nobody wrote down.

GUIDE 58W

At the end of a release the evidence sits in four different states at once. Each state calls for a different sentence. Ask of each claim what would have to happen for it to become settled. Another method? A validated standard? Continued funding? A chemistry trigger? Collapsing four states into one is how the sentence stops being true.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

evidence comes in strengths, and a claim inherits the weakest one under it

VERDICT 86W

At the end of a release the evidence is in several states at once, and each calls for a different recommendation. An identity confirmed by independent methods with controls is established and can be closed. A transformation product that is plausible but has no validated standard is provisional, and saying so is what keeps the record honest. Filtered water meeting criteria over contaminated sediment is a control that

has to keep being funded. And corrosion held inside a narrow chemistry window is conditional operation with triggers.

TAKEAWAY 9W

Chemical readiness is claim-by-claim, not a single green light.

15.2 Audit the electrochemical removal from the charge passed

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · BALLPARK

SCENE 35W

The electrochemical polishing cell has run for 6 hours at 40 amperes. Before the plant trusts the number for the water leaving it, it wants to know how much metal that should have taken out.

GUIDE 69W

Five numbers, and two of them are the same duration in hours and in seconds. Another is the electrons each metal ion needs, which is a property of the metal. Check the units the constant is written in before choosing anything. What comes out is an upper bound rather than a removal. Side reactions carry some of the current. Measured removal below it is efficiency; above it is impossible.

ASK 13W

What upper bound does 6 hours at 40 A place on metal removal?

BEHIND THE BUTTON 119W

What the relationship says. $Q = It$ and $n_{\text{metal}} = Q/(zF) = It/(zF)$.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

current is charge per unit time · a divalent metal ion needs two electrons to become neutral metal

VERDICT 73W

Current times time is charge. Dividing charge by the Faraday constant gives moles of electrons, and a divalent metal needs two electrons per metal ion. 6 hours at 40 A passes 864,000 C, or about 8.95 mol of electrons. The theoretical maximum is therefore about 4.48 mol of metal. Real removal is lower when side reactions carry some current. Comparing measured removal with this bound gives current efficiency and catches impossible treatment claims.

TAKEAWAY 13W

Faraday's law turns measured charge into a quantitative upper bound on electrochemical removal.

15.3 Fund the chemical legacy

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · CHOICE

SCENE 41W

The temporary laboratory is being packed up as the emergency budget closes. The plant is back online, but sediment still holds an estimated 71 kilograms. Four monitoring stations were built during the response; only one has permanent funding after next month.

GUIDE 71W

All four options are defensible uses of the last of the money, and they protect different things. Ask of each what would go unnoticed if it were not funded. Seventy-one kilograms are in sediment, twenty are unaccounted for, and a transformation product is still being characterised. None of that was destroyed; it fell below a threshold. What outlasts a release is the part that went quiet, not the part that stopped.

ASK 13W

The emergency money is about to move elsewhere. What does the city keep?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

falling below a threshold is not the same as being gone

VERDICT 85W

Nothing was destroyed. 71 kilograms are in sediment, and 20 are unaccounted for. A transformation product is also still being characterised. These problems are below a threshold, not gone. Monitoring across water, sediment and air is the only item on the list that would notice any of it coming back. Corrosion control protects the infrastructure, which is a different hazard. Open records preserve the method and

cost almost nothing. What outlasts a release is the part that fell quiet rather than the part that stopped.

TAKEAWAY 9W

Stewardship continues after concentrations fall below the immediate threshold.

15.4 Resolve the ambiguous peak

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM ·
CALLBACK · CHOICE

SCENE 39W

1 peak at 4.2 minutes matches 2 candidates to within 0.05 minutes, and both are on the shortlist. The river sample also carries humic matter, chloride at 40 mg/L, and a detergent from the foam used on the fire.

GUIDE 62W

Two candidates come off the column together. So anything that depends on how fast they travel inherits the ambiguity. Sort the options by that test. Does this one improve the separation, replace it with evidence of another kind, add controls, or only change the presentation? Controls make a number defensible and cannot resolve an overlap. Better presentation adds no measurement at all.

ASK 15W

One peak is still ambiguous and the review board wants it settled. What settles it?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

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TAKES AS READ

two compounds that overlap in one method may not overlap in another

VERDICT 84W

There are two honest ways out of an overlap. Improve the separation until the components stop coming off the column together, or bring in evidence of a different kind so the answer no longer depends on the separation at all. The second is faster, and it cannot inherit the problem: a method that identifies by structure does not care that two compounds travelled at the same speed. The third way is to present the same weak data more attractively, and it adds no measurement.

TAKEAWAY 15W

Better presentation cannot repair a method that was fooled, or evidence that was never validated.

END OF THE DAY 11W

Deliver a claim-by-claim chemical evidence package and a long-term monitoring plan.

THE CLOSING CARD

The plant came back online on day sixty-one, and the city has been drinking its own water since. Every claim in the package held: the compound identified by two methods, the plume mapped, the sediment reservoir bounded, and ten years of monitoring funded before the emergency budget closed.

The fence came down on a Thursday. Both neighbourhoods were home by the weekend, and the river survey still samples the same four stations every month.

Card text only — no options, boards or answers. 13,236 words of card prose across 61 stops. Generated from `themes/contamcity` by `tools/make-cardbook.mjs`.